

# MASSIF & MyceliaLM: Executive Briefing & Funding Request

## Actionable AI Safety, Continuous-Time Telemetry & Ex-Ante Governance

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**Credentials:** NVIDIA Inception Member | Recipient of \$100,000 AWS Cloud Compute Grant

**Traction Note:** Developed & validated self-funded over 12+ months on a single GPU (1.62B parameters)

### Executive Summary & Founder Traction

Standard AI alignment and safety tools operate *post-hoc* and *open-loop*—auditing outputs after model training or inference is complete. The **Multiscale Attractor Stability and Stress Inference Framework (MASSIF)** and **MyceliaLM** (1.62B parameters) introduce **live, continuous-time ex-ante governance** during the forward pass. Built and validated over 12+ months of self-funded research on single-GPU hardware (NVIDIA 16GB T4 / AWS SageMaker), this work demonstrates continuous autonomous governance without human intervention.

### Layer 1: Core Kernel (Plain English Introduction)

**The Problem:** Modern AI training is like watching an athlete run a marathon while only measuring their finish-line time. If the athlete collapses at mile 20, researchers have no early warning, resulting in thousands of dollars in wasted compute. Current models are like bicycles pushed downhill without a rider: they coast until they fall.

**The Solution:** MyceliaLM embeds an **internal nervous system (seismometer)** directly into the transformer’s forward pass. It senses subtle internal vibrations, loss-landscape curvature, and representational drift *while running*, making real-time fractional adjustments to keep the network balanced.

**Decentralized Swarm:** Rather than a monolithic brain moving in lockstep, MyceliaLM desynchronizes its 32 attention heads into a specialized swarm of geometric routers, achieving high efficiency without structural collapse.

### Layer 2: Ethics, Policy & Alignment Vantage Point

**Overcoming Detection Lag:** Evaluating output tokens is fundamentally reactive. Internal representations experience silent *Deep Drift* long before structural corruption reaches the output layer.

**Ex-Ante Interception:** MASSIF replaces post-hoc forensics with live forward-pass governance. Active governors (e.g., *Fibonacci Coherence Attenuation*) damp high-frequency coordinate noise in deeper layers ( $1.618^{-1} \approx 0.618$  Golden Ratio weighting), preventing recursive collapse before it pollutes downstream tokens.

### Layer 3: The Competitive Moat & “Why Now?”

**Competitive Moat:** Current interpretability and safety efforts (e.g., RLHF, post-hoc circuit analysis, output verifiers) are reactive and operate strictly on final model outputs. MASSIF is the first framework to provide **ex-ante, forward-pass governance at the hidden-state level**, catching instabilities layers before they manifest as harmful or hallucinated outputs.

**Why Now? (Agentic Risk):** As AI models transition from single-turn chat to autonomous, multi-step agentic reasoning, internal latent drift accumulates exponentially across reasoning chains. In an agentic world, post-hoc checks occur after an unsafe action has already been taken. The AI industry urgently requires an internal seismometer to govern reasoning *in vivo*.

### Layer 4: Systems, Engineering & Commercial Economics

**Commercial Value:** Autonomous closed-loop governance prevents catastrophic training divergence, saving millions in wasted compute capital. Running on commodity single-GPU hardware (1.62B parameters on 16GB VRAM), MASSIF adds zero latency or memory overhead.

**Closed-Loop P-Controller (SURFER):** SURFER executes real-time target updates ( $\tau_\alpha$ ) based on the Optimization Response Function  $\chi_R = \partial\mathcal{L}/\partial R$ :

$$\tau_\alpha(t+1) = \tau_\alpha(t) \cdot (1 \mp k_p) \quad \text{for } |\chi_R| > 0.05 \quad (k_p = 0.02)$$

### Empirical Audit (145 Telemetry Snapshots, Steps 159k–190k):

- **Kinematic Stability:** Directional SDE SNR  $\rho_{\text{dir}} \approx 100$  (Coefficient of Variation CV = 0.0129).

- **Thermodynamic Volatility:** Optimization response  $CV_{\chi_R} = 2.2592$ .
- **Kinematic-Thermodynamic Decoupling:**  $CV_{\chi_R}/CV_{\rho_{\text{dir}}} = \mathbf{175.12\times}$ , proving optimization can be governed while representational flow remains ultra-stable.
- **Decentralized Flow:** Cross-token coherence  $c_l = 0.000$  alongside high head variance  $\sigma_{\text{head}}^2 \approx 0.730$ .

## Layer 5: Theoretical Physics & Singular Learning Theory

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**Riemannian Trajectories:** Hidden-state trajectories follow low-curvature geodesics ( $\mathbb{E}[\kappa] \approx 0.04$ ) driven by strong logical drift ( $D^{(1)} \approx 19.17$ ) over minimal diffusion ( $D^{(2)} \approx 0.037$ ).

**Singular Variety Mapping:** The order parameter  $R_t = \Pi_\alpha / (\Pi_{\text{FFN}} + \Pi_{\text{MPC}})$  probes transitions in the Real Log Canonical Threshold (RLCT,  $\lambda$ ). Positive zero-crossings ( $\chi_R > 0$ ) mark entry into the **Grokking Window**, shedding redundant toric padding.

## The Ask & Milestones (6-Month Roadmap)

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### Funding Request: \$150,000 for Full-Time Scaling & Open-Source Deployment

To transition MASSIF from a proven 1.62B prototype into an industry-standard AI safety framework, we are seeking **\$150,000** to cover dedicated compute infrastructure and full-time research development over a 6-month horizon.

- **Milestone 1 (Month 2) — 7B Architecture Scaling:** Scale SURFER closed-loop control validation from 1.62B to a 7B parameter transformer architecture, demonstrating multi-GPU telemetry scaling.
- **Milestone 2 (Month 4) — Academic Workshop Publication:** Submit empirical “Kinematic-Thermodynamic Decoupling” results to top-tier AI safety venues (e.g., NeurIPS / ICLR Mechanistic Interpretability workshops).
- **Milestone 3 (Month 6) — Open-Source Release:** Release the full PyTorch MASSIF telemetry hook suite to the developer community, establishing open-source adoption and standardized AI safety observability.